

A Novel SAR Target Recognition Method Combining Electromagnetic Scattering Information and GCN

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Abstract—The existing deep learning studies on synthetic aperture radar (SAR) automatic target recognition (ATR) mainly focus on the utilization of the amplitude of SAR image via convolutional neural network (CNN), while the electromagnetic scattering information is rarely considered. Given that scattering centers (SCs) can characterize the target's electromagnetic scattering characteristics and describe the target's physical structure information, the SC feature should be helpful for SAR ATR. Therefore, we propose a novel SAR ATR method that combines electromagnetic scattering information and graph convolutional network (GCN) effectively and directly. Specifically, we model each extracted SC as a node to convert the SCs into graph data. The constructed graph is learned via GCN to describe the target's physical structure information, where the features of different GCN layers are fused to avoid the oversmoothing of GCN. Label smoothing is combined with GCN for the first time to alleviate the overfitting caused by the limited training data. To the best of our knowledge, this study is the first to introduce the GCN for effectively utilizing the SCs, proving that the structural characteristics of the SCs of SAR targets are remarkably beneficial for recognition. Extensive experimental results on the measured moving and stationary target acquisition and recognition (MSTAR) dataset show that our method can obtain superior recognition performance compared with the existing methods.

Index Terms—Automatic target recognition (ATR), graph convolutional network (GCN), multiscale fusion, scattering center (SC), synthetic aperture radar (SAR).

I. INTRODUCTION

SYNTHETIC aperture radar (SAR) has developed rapidly in recent years due to its all-weather and all-time capabilities and has been widely applied in civilian and military fields. As a key part of intelligent SAR image interpretation, SAR automatic target recognition (ATR) can predict the target categories from SAR images, which has aroused wide attention and has been extensively investigated [1]–[3].

There are some existing SAR ATR methods using the well-designed handcrafted features such as bag-of-visual-word [4], [5] for extracting the features of SAR targets. Using an end-to-end framework, convolutional neural network (CNN) has been the mainstream method for SAR ATR due to its

strong feature representation ability. Chen *et al.* [6] proposed a new all-convolutional network (A-ConvNet) that removes the fully connected architectures in the traditional CNN to reduce training parameters. Amrani and Jiang [7] exploited the extracted deep features from CNNs on SAR images to introduce more powerful discriminative features and robust representation ability. However, these methods only consider the amplitude information of the SAR target, and the electromagnetic scattering characteristic is insufficiently utilized. Some studies have utilized the electromagnetic scattering information to achieve SAR ATR [8], [9]. Zhang *et al.* [8] combined the scattering center (SC) features and Bayesian compressive sensing (BCS) to achieve SAR ATR. Li *et al.* [9] fused the global CNN features with the local component features that are generated via the attributed SCs. However, these methods do not directly combine the electromagnetic scattering parameters with deep neural networks, which are ineffective. Therefore, how to develop a more direct method to integrate the electromagnetic scattering information into deep neural networks is critical for SAR ATR.

Graph convolutional network (GCN) [10] has aroused increasing attention in the last few years due to the strong feature learning and structure learning ability. Compared with the traditional CNN operating convolution on regular Euclidean data such as images and videos, GCN is designed to operate convolution on graph-structured data, such as social networks and biomolecules. Some recent studies [11], [12] have introduced GCN to address few-shot recognition in SAR ATR. Li *et al.* [11] presented a graph meta-learning method for SAR target recognition that requires simulated SAR data to acquire meta-knowledge. Yang *et al.* [12] integrated the GCN into the relation network to measure the similarity among samples for few-shot learning, which only utilizes the label information of the nodes. All these methods model SAR images as nodes to investigate the semantic relationship for node classification. However, the structural relationship implicitly contained in the SAR target is ignored, and the electromagnetic scattering characteristic is not utilized.

Given that SCs can characterize the target's electromagnetic scattering characteristics [8] and describe the target's physical structure information, the SC feature should be helpful for SAR ATR. Considering the advantages of the SC feature, we propose a novel SAR ATR method that combines the targets' electromagnetic scattering information and GCN. We first extract the SCs from the complex SAR image to obtain their physical parameters, which are then used to construct graph data. Inspired by the utilization of multiscale GCN layers [13], the multiscale features generated from different GCN layers are fused to avoid oversmoothing.

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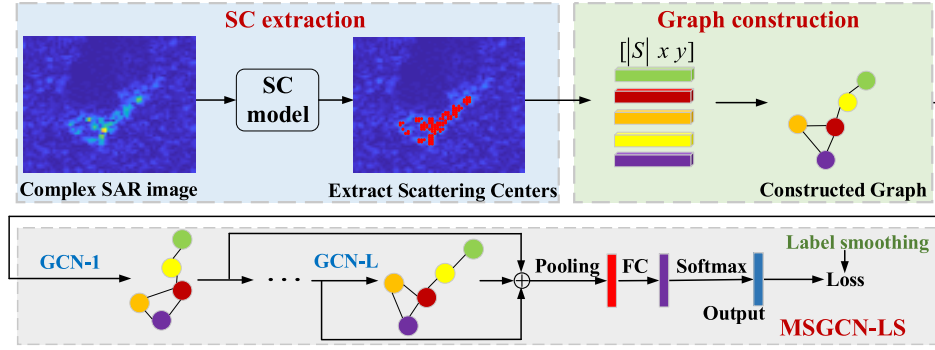


Fig. 1. Flowchart of the proposed method. SCs are first extracted from the complex SAR image and are subsequently constructed into a graph. Our proposed GCN-based network MSGCN-LS is employed for the graph classification.

Label smoothing is combined with GCN for the first time to alleviate the overfitting caused by the limited training data.

Our main contributions can be summarized as follows.

- 1) We are the first to introduce GCN for sufficiently utilizing the SCs, proving that the structural characteristics of the SCs of SAR targets are remarkably beneficial for ATR.
- 2) We directly incorporate electromagnetic scattering information of SAR targets into deep networks, which is more effective than the existing methods.
- 3) Label smoothing is combined with GCN for the first time to alleviate the overfitting caused by the limited training data.

II. PROPOSED TARGET RECOGNITION METHOD

Fig. 1 shows the flowchart of the proposed SAR ATR method. As shown in Fig. 1, our proposed method consists of three stages. First, we extract the SCs from a complex SAR image based on the SC model. Second, the SCs are modeled as a graph, where each node denotes the single SC, and each edge describes the connectivity relationship between nodes. Finally, we use multiscale GCN combined with label smoothing (MSGCN-LS) to realize graph classification. Specifically, we use multiscale GCN to capture the dependencies among nodes in the constructed graph. The structural features are generated via the pooling layer and then fed to the classifier to obtain the prediction result. The label smoothing strategy is combined with multiscale GCN to form the loss function, and the third stage is denoted as MSGCN-LS. The detailed procedures of the three stages are described in Sections II-A–II-C.

A. SC Extraction

The backscattering of a distributed radar object working in a high-frequency region is well approximated as the sum of responses from individual SCs, which can be described as a function of frequency f and aspect angle φ as follows [14]:

$$\mathbf{E}(f, \varphi; \theta) = \sum_{i=1}^K \mathbf{E}_i(f, \varphi; \theta_i) \quad (1)$$

where K denotes the total number of the SCs and $\theta = \{\theta_i\} = [S_i, x_i, y_i]$ represents the SCs' parameter set. The backscattering field of an individual SC can be described by using the SC model as follows:

$$\mathbf{E}_i(f, \varphi; \theta_i) = S_i \cdot \exp\left(\frac{-j4\pi f}{c}(x_i \cos \varphi + y_i \sin \varphi)\right) \quad (2)$$

where c represents the speed of the electromagnetic wave; for the i th SC, S_i denotes the complex amplitude, and x_i and y_i represent the position coordinates along the range dimension and azimuth dimension, respectively.

The SC model provides a physically relevant description of the target by using a set of physical parameters, including positions and amplitudes. The parameter estimation is achieved based on sparse representation via the orthogonal matching pursuit (OMP) [15] algorithm.

B. Graph Construction

Based on the SC model, each SC is associated with an amplitude S_i and a spatial position $\{x_i, y_i\}$. A graph $G = (V, E)$ is constructed based on the extracted SCs to learn the target's physical structure information. V and E denote the set of nodes and edges of the graph, respectively. Each node denotes an SC and the edges represent the relationship between the nodes. We use the adjacency matrix $A \in R^{n \times n}$ to describe the edge information. The initial features of nodes V are obtained by using the parameters $[|S|xy]$ extracted by the SC model, which is defined as $H \in R^{n \times d}$. Here, n denotes the number of nodes and d denotes the input feature dimension.

The relationship among SCs is assumed to be related to their spatial positions. Specifically, the closer the SCs in the spatial positions, the stronger the connectivity relationship between nodes. Based on this assumption, the adjacency matrix is defined by the distance between nodes and normalized with Gaussian kernel [16], which is formulated as

$$a_{i,j} = \begin{cases} e^{\frac{\|m_i - m_j\|^2}{-2\gamma^2}}, & \text{if } \|m_i - m_j\|^2 < \tau \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $m_i = [x_i y_i]$, parameter γ denotes the standard deviation of the Gaussian kernel, and τ denotes the distance threshold. When the Euclidean distance between two nodes is less than τ , the greater γ leads to the weaker dependency between nodes.

C. Multiscale GCN Combined With Label Smoothing

Given that the constructed graph is graph-structured data, the GCN can be utilized to capture the dependencies among SCs. Analogous to the stacking of multiple layers in CNN, multiple GCN layers are stacked in our GCN-based model. The graph features and the adjacency matrix are simultaneously taken as the input of the GCN layer. Mathematically, the output graph feature of the first GCN layer can be represented

as

$$h^{(1)} = \sigma(AH^{(0)}) \quad (4)$$

where σ denotes the nonlinear activation function and rectified linear unit (ReLU) is used in our method, A is the normalized adjacency matrix, and $W^{(0)}$ denotes the weight matrix of the first GCN layer. Analogous to stacking multiple CNN layers, higher order features can be aggregated from neighbors by stacking multiple GCN layers as follows:

$$h^{(l+1)} = \sigma(Ah^{(l)}W^{(l)}) \quad (5)$$

where l denotes the number of GCN layers, $h^{(0)} = H$, and $W^{(l)}$ denotes the weight matrix of the l th layer. To learn the different fields of neighboring information, we fuse the node features in different GCN layers via the concatenation operation along the feature dimension, which can be denoted as $h^{\text{fus}} = [h^{(1)}, \dots, h^{(L_{\text{GCN}})}]$, where L_{GCN} denotes the number of GCN layers. A simple global average-pooling operation is employed on the fused graph features to generate the global structural feature h_{global} , which is subsequently fed into the fully connected layer and output the recognition probability via softmax, which can be formulated as follows:

$$\hat{y} = \text{softmax}(W_{\text{fc}}h_{\text{global}} + b) \quad (6)$$

where W_{fc} and b denote the weight matrix and the bias vector of the fully connected layer, respectively.

Label smoothing [18] is proposed in GoogleNet as model regularization to prevent overfitting. For a training sample with ground-truth label y , the mixed distribution used in label smoothing is presented as follows:

$$y^{\text{LS}} = (1 - \varepsilon)y_{\text{one-hot}} + \varepsilon/T \quad (7)$$

where ε is a smoothing parameter, $y_{\text{one-hot}}$ is the one-hot vector of ground-truth label y , and T denotes the class number of training data. Therefore, the final loss function of MSGCN-LS is represented as follows:

$$L_{\text{MSGCN-LS}} = -\frac{1}{2N} \sum_{i=1}^N y_i^{\text{LS}} \log(\hat{y}_i) \quad (8)$$

where N denotes the number of samples in the training set and $\hat{y}_i$ is the output probability from the model.

D. Training Procedure

The detailed training procedure of our proposed method is described in Algorithm 1.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Dataset Description

The most widely used SAR ATR dataset, moving and stationary target acquisition and recognition (MSTAR) dataset, which includes ten different types of ground targets, is used in the experiments. The MSTAR dataset includes two commonly used experimental settings, namely, the three-target data and the ten-target data. The three-target data includes one type of armored vehicle BTR70 and two types of tanks BMP2 and T72. Another seven ground vehicles are contained in the ten-target data. The detailed information of the three-target data and ten-target data [2], [3], [9] is listed in Tables I and II, respectively.

Algorithm 1 Training Procedure of the Proposed Method

Input: Training set, K , ε , τ , γ , L_{GCN} .

Output: Parameters $\Phi = \{W^{(0)}, \dots, W^{(L)}, W_{\text{fc}}, b\}$.

- 1: Initialize the parameter Φ .
- 2: Extract the SCs by using the OMP algorithm [15].
- 3: Construct the graph in accordance with (3).
- 4: **While** not converged **do**
- 5: Generate the multi-scale GCN features by using (5).
- 6: Fuse the multi-scale GCN features via concatenation operation.
- 7: Obtain the prediction probability from (6).
- 8: Optimize $L_{\text{MSGCN-LS}}$ with Adam from (8)
- 9: **end for**

TABLE I
DETAILED INFORMATION OF THE THREE-TARGET DATA

Dataset	BMP2			BTR70		T72	
	C21	9566	9563	C71	132	S7	812
Training (17°)	233	0	0	233	232	0	0
Test (15°)	196	196	195	196	196	191	195

B. Experiment Setup

In the MSTAR experiments, the original SAR images are center cropped into 64×64 to reduce the computation and storage cost. We extract 400 SCs for each complex-valued SAR image by using the OMP algorithm. L_{GCN} is set to 3, and the output feature dimension of each GCN layer is set to 512. The initial learning rate with Adam is set to 0.001 and decays by 0.50 every 100 epochs. The total training epochs are set to 500. The label smoothing hyperparameter ε is set to 0.05.

C. Three-Target MSTAR Data Experiments

In this section, we evaluate the performance of our proposed method with several traditional SAR ATR methods and deep learning-based methods. The traditional methods used here include support vector machine (SVM), principal component analysis (PCA), template matching, dictionary learning and joint dynamic sparse representation (DL-JDSR) [2], BCS [8], and SAR-oriented GBVS [19]. The deep learning-based methods used here include the Euclidean distance restricted auto-encoder (Euclidean-AE) [3], A-ConvNet [6], Visual Geometry Group CNN (VGGNet) [18], multiscale CNN based on component analysis (CA-MCNN) [9], Yolo-MCCA [20], and deep feature extraction [7]. BCS and CA-MCNN utilize electromagnetic scattering information to achieve robust recognition results. The GCN-based methods illustrated in the introduction of SAR ATR consider the few-shot learning tasks, which cannot be compared with our recognition task. All these comparative methods do not adopt any data augmentation for fairness. Tables III and IV show the recognition results of the proposed method and the comparison methods.

In Tables III and IV, our proposed method has better accuracy compared with other methods. Although CA-MCNN fuses CNN features and SC features for ATR, our method still achieves higher accuracy by only using SC features. Compared with BCS, the accuracy of our method has an

TABLE II
DETAILED INFORMATION OF THE TEN-TARGET DATA

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

TABLE III
DETAILED RECOGNITION RESULTS OF EACH CATEGORY ON THE THREE-TARGET DATA

	BMP2	BTR70	T72	Accuracy
Proposed method	0.9847	0.9949	0.9880	0.9875
CA-MCNN [9]	0.9813	0.9949	0.9880	0.9861
BCS [8]	0.9490	1	0.9870	0.9790
SVM	0.7325	0.9643	0.9278	0.8491
PCA	0.8330	0.9541	0.9106	0.8835
Template matching	0.9148	1	0.9244	0.9311
DL-JDSR [2]	0.9301	0.9898	0.9312	0.9391
Euclidean-AE [3]	0.9421	0.9388	0.9416	0.9414
VGGNet	0.9642	0.9694	0.9485	0.9582
A-ConvNet	0.9676	0.9949	0.9605	0.9684

TABLE IV
COMPARISON RESULTS OF OUR PROPOSED METHOD WITH SAR-ORIENTED GBVS ON THE THREE TARGETS [19]

	BMP2	BTR70	T72	Accuracy
Proposed method	1	1	1	1
SAR-Oriented GBVS [19]	0.9964	0.9964	1	0.9976

TABLE V
ABLATION EXPERIMENTS

Single-layer($L_{GCN}=1$)	Multi scale fusion	Label smoothing	Accuracy
✓	×	✓	0.9429
×	×	✓	0.9824
×	✓	×	0.9810
×	✓	✓	0.9875

advantage of 0.85%, indicating that the utilization of SC features of our proposed method is more effective. Compared with other deep learning-based methods such as VGGNet and A-ConvNet, our method has higher accuracy. From Table III, our proposed method obtains the highest accuracy on types BMP2 and T72, which contain variants, and other methods except for CA-MCNN perform worse on the two types. The above experimental results on the three-target data reveal that our proposed method can extract a more discriminative feature representation.

D. Ablation Experiment

The ablation experiments are given in Table V to analyze the effects of stacking GCN layers, multiscale fusion, and label smoothing.

From Table V, the performance is worse than the multilayer case when $L_{GCN} = 1$, indicating that stacking GCN layers is beneficial. Comparing rows 3 and 4, the accuracy can be increased by 0.65% by adding label smoothing into our model. This condition may be because the MSTAR dataset is limited for training GCNs. Thus, the addition of label smoothing into

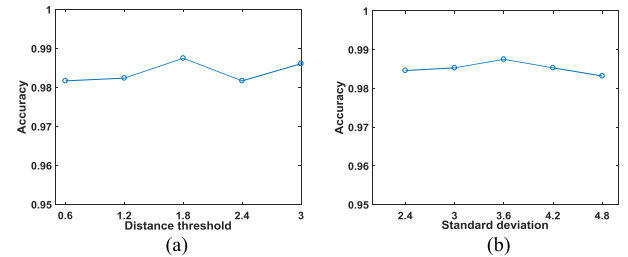


Fig. 2. Accuracies with different thresholds and standard deviations. (a) Different thresholds. (b) Different standard deviations.

our method can alleviate overfitting to some extent. From rows 2 and 4, the utilization of multiscale fusion can slightly improve the accuracy, thereby demonstrating the advantages of multiscale fusion.

E. Model Analysis

The recognition experiments with different hyperparameters on the three-target data are performed to analyze the effects of different distance thresholds and standard deviations in the graph construction stage on the model performance. The results are shown in Fig. 2.

1) *Effect of Threshold*: As shown in Fig. 2(a), we can see that the recognition performance varies with different distance thresholds. The optimal value of the distance threshold is 1.8. When τ exceeds or less than 1.8, the performance obviously dropped, indicating that the constructed graph is extremely dense or extremely sparse to represent the structure of targets.

2) *Effect of Gaussian Standard Deviation*: From Fig. 2(b), the optimal value of standard deviation γ is 3.6. The slight changes in Fig. 2(b) indicate that our proposed method is insensitive to the value of standard deviation.

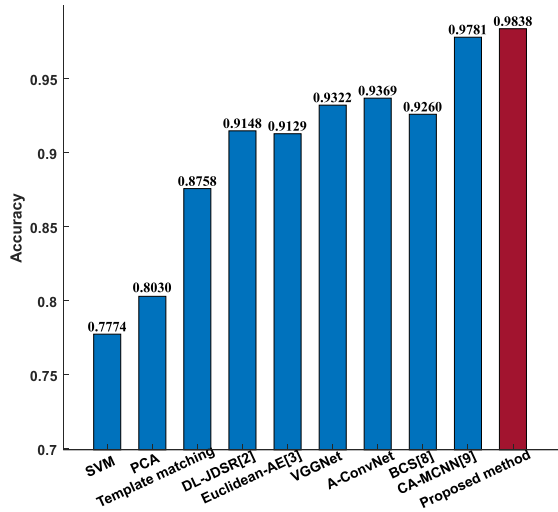


Fig. 3. Accuracies of different SAR ATR methods on the ten-target data.

TABLE VI

COMPARISON RESULTS OF OUR PROPOSED METHOD WITH SAR-ORIENTED GBVS, YOLO-MCCA, AND DEEP FEATURE EXTRACTION ON THE TEN TARGETS [7], [19], [20]

	Accuracy
Proposed method	0.9984
SAR-Oriented GBVS [19]	0.9968
Yolo-MCCA [20]	0.9970
Deep feature extraction [7]	0.9982

F. Ten-Target Experiments

In this section, the MSTAR ten-target data are used to evaluate the recognition performance of our proposed method with other comparison methods. The comparison methods are the same as those in the three-target experiments, and the recognition results are shown in Fig. 3 and Table VI.

As shown in Fig. 3 and Table VI, our proposed method obtains higher accuracy than all the comparison methods. Compared with CA-MCNN, our method obtains 0.57% higher accuracy by only using SCs features. Compared with other deep learning-based methods, our proposed method has an obvious advantage. The recognition results on the ten-target data demonstrate that our proposed method has superior recognition performance with more target types.

IV. CONCLUSION

For exploiting the information of SAR targets more efficiently, a novel recognition method is proposed to efficiently utilize the electromagnetic scattering information for SAR ATR. Each extracted SC is modeled as a node to convert the SCs into graph-structured data, which can be effectively learned via MSGCN-LS. To the best of our knowledge, this study is the first to introduce GCN for effectively utilizing the SCs and learning the target's physical structure information for SAR ATR. The extensive experimental results on the measured MSTAR dataset illustrate that our method can obtain superior recognition performance compared with the existing methods. Our future work will focus on the dynamic learning of GCN to enhance the adaptative ability to various SAR targets under

open set condition [21] and improve the robustness of our proposed method under the extended operating conditions for SAR ATR.

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